Notes: Pierce and Schott (AER: Insights, 2020)

Trade Liberalization and Mortality: Evidence from US Counties

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1 The question and the empirical logic

The paper asks whether a large and persistent economic shock—a major episode of US trade liberalization toward China—causally affects US mortality, with a focus on "deaths of despair" (drug overdose, suicide, and alcohol-related liver disease, ARLD). The key empirical challenge is to separate the effect of local labor-market exposure to trade from confounding local trends and national shocks. The authors exploit a plausibly exogenous and pre-determined measure of exposure to *Permanent Normal Trade Relations* (PNTR) to China (October 2000): counties that specialized in industries facing larger potential tariff increases *absent* annual renewal experienced a larger effective liberalization when PNTR made low tariffs permanent. They implement a generalized difference-in-differences / event-study design using county-by-year mortality data (1990–2013), with extensive controls and fixed effects.

Main takeaway: counties more exposed to PNTR experience relative increases in deaths of despair, and this is driven primarily by fatal drug overdoses among working-age whites; the timing aligns with deterioration in local labor-market conditions and increased disability uptake.

2 Institutional background: what PNTR changed

Before PNTR, China generally faced low *Normal Trade Relations* (NTR, also called MFN) tariff rates, but continued access to these low rates required annual renewal under US law. If renewal failed, tariffs on Chinese imports would jump to much higher *non-NTR* (Smoot–Hawley schedule) rates. This renewal risk created trade-policy uncertainty that discouraged some trade and investment relationships. PNTR (October 2000, implemented around China's WTO accession) eliminated annual renewals, making low NTR tariffs permanent and thereby increasing incentives for production in China and exports to the US. The economic force emphasized in the paper is a *differential* increase in import competition across US locations, driven by their pre-policy industry composition.

3 Data and core outcome construction

3.1 Mortality data and outcomes

The authors use death-certificate microdata from the CDC (1990–2013), aggregated to the county-year level and disaggregated by: age bins, gender, race, and cause of death. They construct (i) crude and (ii) age-adjusted death rates, expressed per 100,000 population, using county-year population denominators from SEER/NCI. The baseline dependent variable in the main event-study regressions is the **age-adjusted** death rate for a specific cause (drug overdose, suicide, ARLD) or for their sum (deaths of despair).

3.2 Why age adjustment matters

Age adjustment removes changes in mortality mechanically due to shifts in age composition. Because counties differ in aging and migration patterns, age adjustment increases comparability across space and time, and it is a standard choice in mortality economics.

4 Exposure to PNTR: the key treatment variable

4.1 Industry-level exposure: the NTR gap

Let j index industries (SIC). Let:

- $NonNTRRate_j$ denote the non-NTR ad valorem tariff rate (Smoot–Hawley schedule, set in 1930),
- $NTRRate_j$ denote the applied NTR tariff rate in 1999 (pre-PNTR).

The industry's exposure to PNTR is the *NTR gap*:

$$NTRGap_j \;=\; NonNTRRate_j - NTRRate_j. \tag{1}$$

Interpretation. $NTRGap_j$ measures the size of the tariff jump that would occur for industry j if annual renewal failed. A larger gap implies more trade-policy uncertainty pre-2000 and therefore a larger effective liberalization when PNTR removed the renewal risk.

Why it is plausibly exogenous. A key point is that most cross-industry variation in $NTRGap_j$ comes from $NonNTRRate_j$, which was set decades earlier (1930) and is not chosen in response to 1990s–2000s US labor-market conditions. This reduces concerns about reverse causality from contemporaneous economic trends to tariff-policy measures.

4.2 County-level exposure: shift-share using pre-period employment

Let c index counties and let L_{jc}^{1990} denote 1990 county employment in industry j. Let $L_c^{1990} = \sum_j L_{jc}^{1990}$ denote total county employment. County exposure to PNTR is the 1990 employment-share-weighted average NTR gap:

$$NTRGap_c \;=\; \sum_j \frac{L_{jc}^{1990}}{L_c^{1990}}\, NTRGap_j. \tag{2}$$

Interpretation (shift-share). Counties with larger pre-2000 employment shares in "high-gap" industries are predicted to face larger post-2000 increases in Chinese import competition (and broader trade adjustment). Using 1990 shares makes exposure predetermined with respect to the policy change in 2000.

Measurement note. Industries not subject to tariffs (e.g., many services) receive $NTRGap_j = 0$. In practice, this concentrates identifying variation in tradable sectors (manufacturing and agriculture).

5 Empirical model: generalized DID / event-study specification

5.1 Baseline regression

Let $DeathRate_{ct}$ denote an age-adjusted death rate (per 100,000) in county c and year t for a specified cause of death. The baseline event-study DID is:

$$
\begin{aligned}
DeathRate_{ct} = &\sum_t \theta_t \, \mathbf{1}\{year = t\} \times NTRGap_c \; + \; \beta X_{ct} \\
&+ \sum_t \gamma_t \, \mathbf{1}\{year = t\} \times X_c \; + \; \delta_c + \delta_t + \varepsilon_{ct}.
\end{aligned} \tag{3}
$$

5.2 Objects, controls, and fixed effects

- θ_t is the key object: a time profile of the association between exposure and mortality relative to the omitted base year (1990).
- δ_c are county fixed effects (absorb time-invariant county differences).
- δ_t are year fixed effects (absorb national shocks common to all counties).
- X_{ct} are time-varying controls capturing other policy changes (e.g., US tariff levels, MFA phase-out exposure, China-side policy changes).
- X_c are baseline (1990) county characteristics (e.g., income, education, veteran share, foreign-born share, manufacturing share), interacted with year dummies so their relationship with mortality can vary flexibly over time.

5.3 Estimation details

Regressions are population-weighted (1990 population), and standard errors are clustered at the state level. The sample spans 1990–2013.

5.4 How to read θ_t: pre-trends and post effects

- **Pre-trend check:** if $\theta_t \approx 0$ for $t < 2000$, then high- and low-exposure counties exhibit parallel mortality trends prior to PNTR.
- **Treatment timing:** a visible break after 2000 supports a causal interpretation tied to PNTR rather than slow-moving confounders.

5.5 Economic magnitude: interquartile shift

The figures translate θ_t into an economically meaningful effect by multiplying coefficients by an interquartile shift in exposure:

$$\Delta^{IQR} NTRGap_c := NTRGap_{c,75th} - NTRGap_{c,25th}.$$

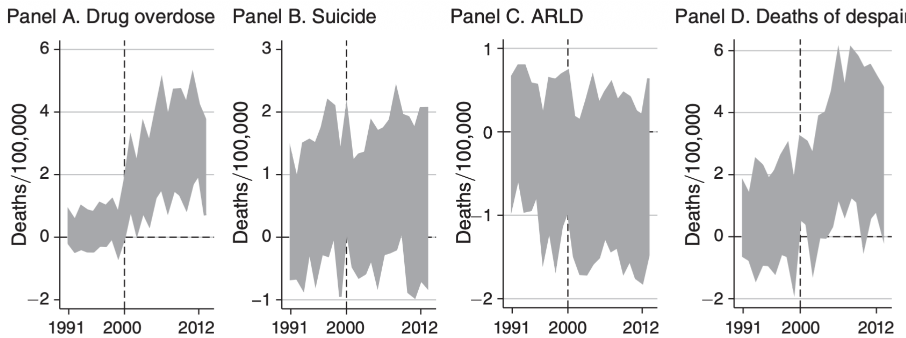

FIGURE 1. IMPLIED IMPACT OF PNTR ON DEATHS OF DESPAIR

The implied impact in year t is:

$$\Delta \widehat{DeathRate}_t = \widehat{\theta}_t \cdot \Delta^{IQR} NTRGap_c.$$

Figures plot these implied impacts with 95% confidence intervals.

6 Core empirical findings (Figures 1–3): mortality responses

6.1 Figure 1: which deaths of despair respond?

Figure 1 reports the implied impact of an interquartile shift in $NTRGap_c$ on:

- Panel A: drug overdoses,
- Panel B: suicide,
- Panel C: ARLD,
- Panel D: overall deaths of despair (sum of the above).

Interpretation. The dynamic profile in Panel A shows:

- no differential trend before 2000 (confidence intervals include zero),
- a clear upward shift after 2000, consistent with PNTR timing,
- economically meaningful magnitudes (roughly a few deaths per 100,000 in the years after PNTR).

Panels B and C show little evidence of a systematic relationship for suicide or ARLD. Panel D rises because overdoses are quantitatively large relative to other components and dominate the sum.

Substantive message. Trade exposure affects deaths of despair mainly through *drug overdoses*, not through suicide or liver disease in this setting.

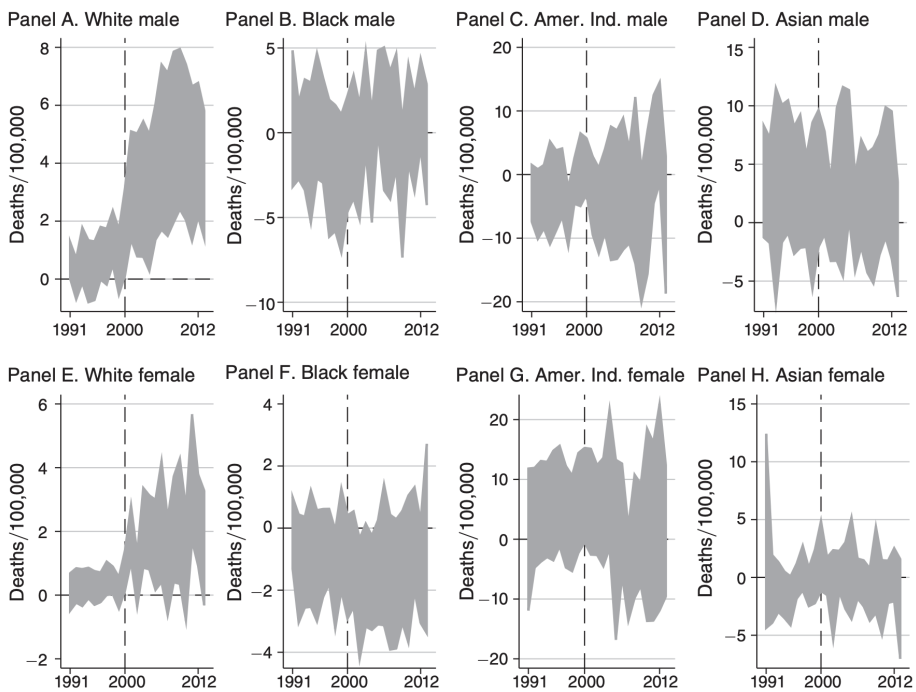

FIGURE 2. IMPLIED IMPACT OF PNTR ON DRUG OVERDOSE DEATHS, BY GENDER AND RACE

6.2 Figure 2: heterogeneity by race and gender

Figure 2 repeats the baseline specification for drug overdoses separately by gender and race. The key visual pattern is that the positive post-2000 response is concentrated among **whites**, especially white males, with white females also showing an increase. Other racial groups display weak or statistically imprecise responses, consistent with differences in baseline labor-market exposure and/or sample size (noisy mortality in smaller populations).

Economic interpretation. A natural interpretation, consistent with the paper's discussion, is that PNTR-driven labor-market disruption was more intense for demographic groups overrepresented in affected manufacturing occupations in the pre-2000 period.

6.3 Figure 3: heterogeneity by age (for whites)

Figure 3 focuses on whites and estimates the PNTR–overdose link by five-year age bins spanning ages 20–64. The figure indicates that the relationship is strongest across much of working age (especially 20–54), and weaker/noisier at older working ages. This age profile strengthens a labor-market mechanism interpretation: the largest responses are among groups most attached to local employment conditions.

7 Robustness and alternative explanations (Figure 4)

Figure 4 assesses whether the overdose result is sensitive to additional controls that target alternative channels:

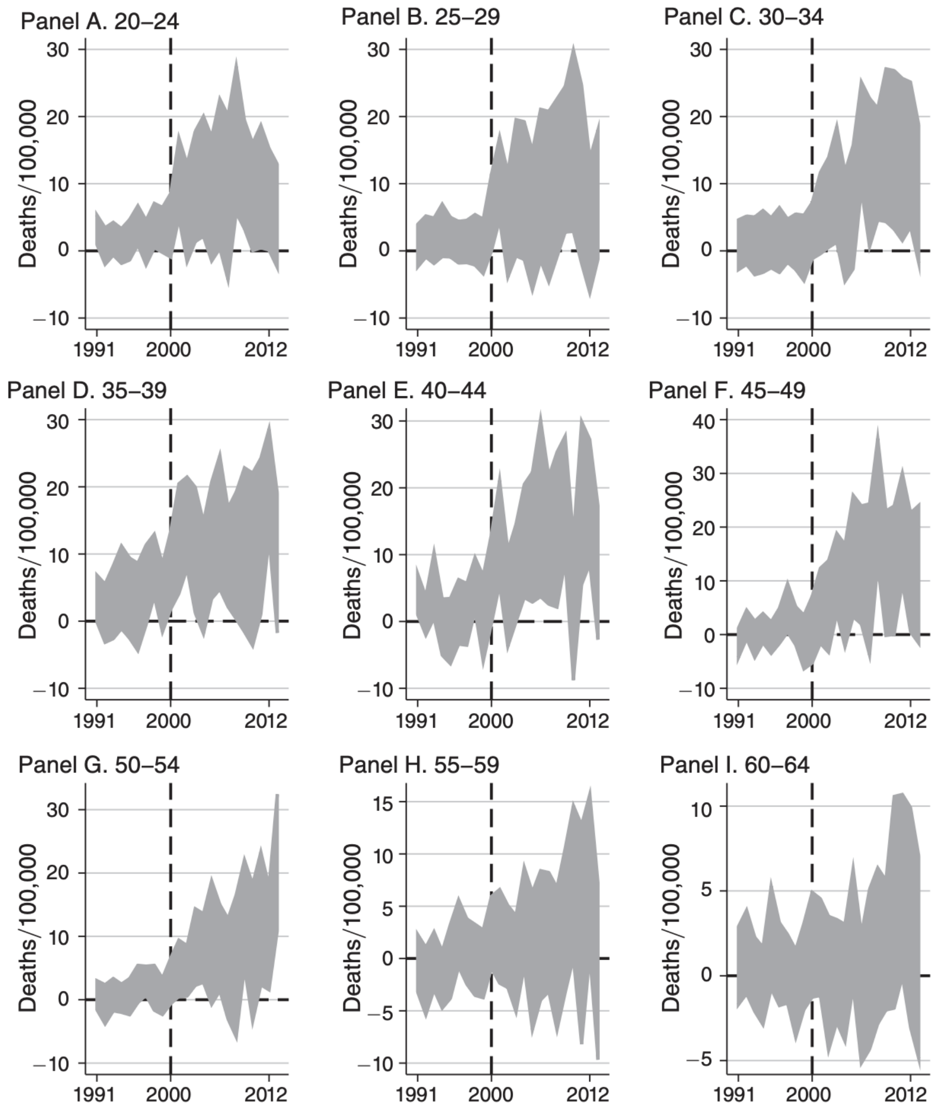

FIGURE 3. IMPLIED IMPACT OF PNTR ON WHITE DRUG OVERDOSE DEATHS, BY AGE

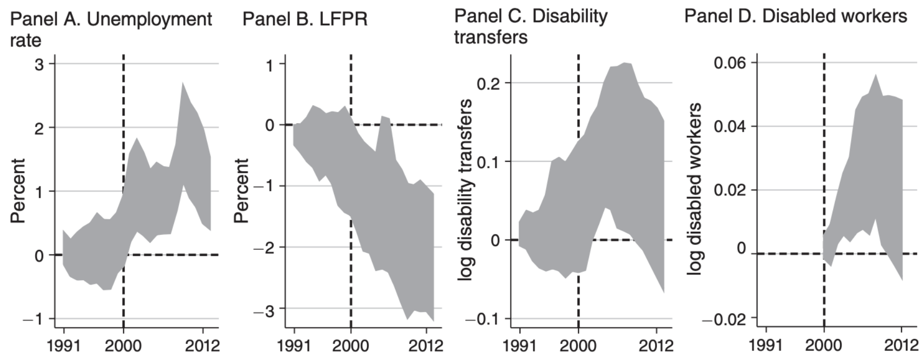

FIGURE 5. IMPLIED IMPACT OF PNTR ON LABOR MARKET OUTCOMES

- Panel A: baseline result (from Figure 1, overdose panel),
- Panel B: controls for state-year Medicaid expansions (healthcare access),
- Panel C: controls for state opioid-law restrictiveness (opioid supply/regulation),
- Panel D: state-by-year fixed effects (absorbing any state-year shocks and policies).

Interpretation. The overdose effect remains qualitatively similar with Medicaid and opioid-law controls. Including state-by-year fixed effects substantially reduces identifying variation (because the treatment varies largely across counties within states), so confidence intervals widen, but the estimated pattern still shifts upward post-2000.

What this rules out (and what it does not). These exercises reduce concern that the main result is mechanically driven by:

- state-level health policy coinciding with PNTR, or
- state-level opioid regulatory changes correlated with exposure.

They do not fully rule out all endogenous opioid supply responses at finer geographic levels, but they make a purely policy-driven explanation less plausible.

8 Mechanisms: labor markets and disability (Figure 5)

Figure 5 uses the same event-study specification to examine outcomes consistent with a labor-market channel:

- Panel A: unemployment rate,
- Panel B: labor force participation rate (LFPR),
- Panel C: log disability transfers,
- Panel D: log number of disabled workers.

Interpretation. Counties with greater PNTR exposure experience:

(i) rising unemployment after 2000,

(ii) falling labor force participation after 2000,

(iii) increases in disability transfers and disability recipients after 2000.

These time patterns align with the timing of the overdose mortality response. This evidence is not a definitive mediation decomposition, but it is consistent with a narrative in which PNTR increases economic distress and weakens labor-market opportunities, which in turn may increase susceptibility to substance abuse and overdose risk.

9 Models and identifying assumptions: what must be true for a causal reading

9.1 Parallel trends in the event-study sense

A causal DID interpretation requires that, absent PNTR, high- and low-exposure counties would have followed similar mortality trends after conditioning on the included controls and fixed effects. Empirically, the paper provides support via:

- near-zero pre-2000 coefficients in Figures 1–5,
- rich controls (including baseline-characteristic-by-year interactions),
- robustness to alternative policies and fixed effects.

9.2 Exogeneity of exposure measure

The NTR gap is largely driven by the non-NTR schedule set in 1930, and county exposure uses pre-policy (1990) industry composition. This reduces the scope for contemporaneous confounding, though the shift-share structure still invites the usual caution: if counties specialized in high-gap industries were on systematically different trajectories for reasons not captured by the controls, estimates could be biased. The paper mitigates this risk with flexible controls and pre-trend evidence.

9.3 Interpretation of estimates

The coefficients are best read as *reduced-form impacts* of a trade-policy liberalization shock on mortality and related outcomes. They do not deliver a structural model of drug markets or health behavior, but they establish a robust and policy-relevant association with strong timing evidence.

10 Conclusion and intuition

10.1 Coherent summary: what the paper concludes

This paper links a large, persistent, and plausibly exogenous change in US trade policy—the granting of Permanent Normal Trade Relations (PNTR) to China in October 2000—to subsequent changes in mortality outcomes across US counties. Using county-level exposure to PNTR constructed from pre-policy industry composition and tariff "NTR gaps," and using mortality microdata for 1990–2013, the authors estimate generalized difference-in-differences/event-study specifications.

The central conclusion is that counties more exposed to PNTR experience *relative increases in deaths of despair*, and that this relationship is driven primarily by **fatal drug overdoses among whites in the working-age population**. The paper does *not* claim that PNTR is the sole driver of the broader rise in drug mortality; rather, it documents a statistically and economically meaningful contribution of a major labor-market shock to a specific component of the "deaths of despair" phenomenon.

10.2 Main empirical takeaways

- **Timing (no pre-trends):** The rise in overdose mortality in more-exposed counties occurs around the time of PNTR and is not explained by differential pre-2000 trends in mortality.

- **Who is affected:** The estimated relationship is concentrated among **whites**, particularly in working-age bins (the paper reports results for age bins spanning the working-age range).

- **Which causes of death:** Within "deaths of despair," the association is **driven by drug overdoses**. In contrast, the paper finds **little relationship** between PNTR exposure and mortality from *suicide* or *alcohol-related liver disease*.

- **Magnitude:** Coefficients imply that an *interquartile shift* in county exposure is associated with a **relative increase** in deaths of despair of roughly **2–3 deaths per 100,000** (about **10–15%** of the average deaths-of-despair mortality rate across counties in 2000), and a similarly sized **2–3 per 100,000** relative increase in **drug overdose** mortality (a sizable share of the roughly **5 per 100,000** average overdose death rate across counties in 2000).

- **Robustness to opioid-policy confounds:** The results are robust to controlling for state-level policy changes related to opioid availability and health care, alleviating (though not eliminating) concerns that the estimates are mechanically driven by contemporaneous opioid-supply regulation.

10.3 Intuition: why would a trade shock raise overdose mortality?

The paper's mechanism story can be understood as a labor-market-distress channel that operates through geographically concentrated shocks:

1. **PNTR is a persistent and uneven local shock.** Because counties differ in their pre-2000 industry structure, PNTR differentially increases exposure to import competition across places. This generates large, long-lived, and spatially concentrated disruptions to local labor markets.

2. **Labor-market deterioration can increase "deaths of despair."** If a local shock reduces employment opportunities, earnings trajectories, and perceived economic prospects, it can increase stress and substance use and can weaken social/economic buffers that otherwise reduce drug abuse and overdose risk. The key point is not that the trade shock changes biology directly; it changes the economic environment in which risky behaviors and health outcomes are produced.

3. **Why the paper emphasizes overdoses.** Drug overdose mortality is plausibly more sensitive to short- and medium-run changes in economic conditions (especially in places where drug supply is already present) than other causes of death. The paper's results are consistent with overdoses being the margin of "deaths of despair" most responsive to this type of labor-market disruption.

4. **Why the demographic concentration matters.** The empirical concentration among whites is consistent with the broader "deaths of despair" pattern documented in prior work and suggests heterogeneous vulnerability (or exposure) across groups. The paper's contribution is to show that a major labor-market shock can map into mortality outcomes in a demographically uneven way.

10.4 Supporting evidence on mechanisms (what the paper shows)

To connect the mortality results to economic channels, the paper documents that counties more exposed to PNTR also exhibit:

- adverse relative changes in labor-market conditions (e.g., unemployment and labor-force participation patterns consistent with weaker local opportunities), and
- increased disability insurance uptake and disability-related transfers.

This evidence supports the interpretation that the PNTR shock operates through economic distress and reduced labor-market attachment, rather than through purely medical-policy channels.

10.5 Interpretation and scope (what the paper does *not* claim)

- The estimates should not be interpreted as a complete welfare accounting of PNTR. The paper focuses on distributional consequences in health and mortality, not on aggregate gains from trade.
- The paper does not claim that trade policy alone "caused" the opioid crisis. Instead, it provides evidence that a major labor-market shock is one meaningful contributor to the observed rise in overdose deaths in more-exposed areas.

10.6 One-sentence takeaway

Counties more exposed to the PNTR trade liberalization experienced a post-2000 relative rise in deaths of despair driven by fatal drug overdoses among working-age whites, consistent with a labor-market-disruption mechanism rather than pre-trends or contemporaneous opioid-policy changes.